

WEEK 4 REPORT-Urekha V

Exploratory Data Analysis of Shopify Stock Data Using Power BI

Dataset: Shopify Stock Data

Tool Used: Microsoft Power BI Desktop

1. Introduction

This project focuses on performing exploratory data analysis on Shopify stock market data using Microsoft Power BI.

The analysis focuses on understanding stock price movements, trading volume, moving averages and market trends over time. Power BI was used to create DAX measures, moving averages, crossover indicators, interactive visualizations, slicers and a metric selector.

The final dashboard provides an interactive view of Shopify stock performance and allows different financial metrics to be analyzed.

2. Objectives

The main objectives of this project are:

- To perform exploratory data analysis on Shopify stock data.
- To calculate 20-day and 50-day Simple Moving Averages.
- To identify bullish and bearish moving-average conditions.
- To analyze stock price trends over time.
- To analyze trading volume and volume trends.
- To create interactive date-based filters.
- To create a metric selector using Field Parameters.
- To create interactive Power BI visualizations.
- To build a dashboard for Shopify stock analysis.

3. Dataset Description

The dataset used for this project is **Shopify Stock Data**.

The major fields used for analysis include:

Field	Description
Date	Date of the stock record
Open	Opening stock price
High	Highest stock price
Low	Lowest stock price
Close	Closing stock price
Volume	Number of shares traded

The dataset was previously prepared and connected with the Dim_Date table from the Week 3 Power BI project.

4. KPI Cards

Five KPI cards were created to provide a quick summary of the Shopify stock data.

The cards display:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

These cards allow the user to quickly understand the main stock values without examining the complete dataset.

Open Price 48.46	High Price 176.29	Low Price 1.85
Close Price 48.46	Trading Volume 41bn	

5. 20-Day Simple Moving Average

A **20-Day Simple Moving Average (SMA)** was created to analyze the short-term trend of trading volume.

The `DATESINPERIOD()` function was used to calculate the moving average over a 20-day period.

The measure was created as:

```

Volume 20 SMA =
VAR CurrentDate = MAX(Dim_Date[Date])
VAR PeriodDays = 20
VAR DateInPeriod =
    DATESINPERIOD(
        Dim_Date[Date],
        CurrentDate,
        -PeriodDays,
        DAY
    )
RETURN
    CALCULATE(
        AVERAGE(Stock_Data[volume]),
        DateInPeriod
    )

```

The 20-Day SMA helps smooth daily fluctuations and provides a clearer view of the short-term trading-volume trend.

6. 50-Day Simple Moving Average

A **50-Day Simple Moving Average** was also created to represent a longer-term trend.

The 50-day moving average helps compare the longer-term trading-volume pattern with the shorter 20-day trend.

The measure was created using DATESINPERIOD() with a 50-day period

Volume 50 SMA =

VAR CurrentDate = MAX(Dim_Date[Date])

VAR PeriodDays = 50

VAR DateInPeriod =

DATESINPERIOD(

Dim_Date[Date],

CurrentDate,

-PeriodDays,

DAY

)

RETURN

CALCULATE(

AVERAGE(Stock_Data[volume]),

DateInPeriod

)

7. Price Trend Analysis

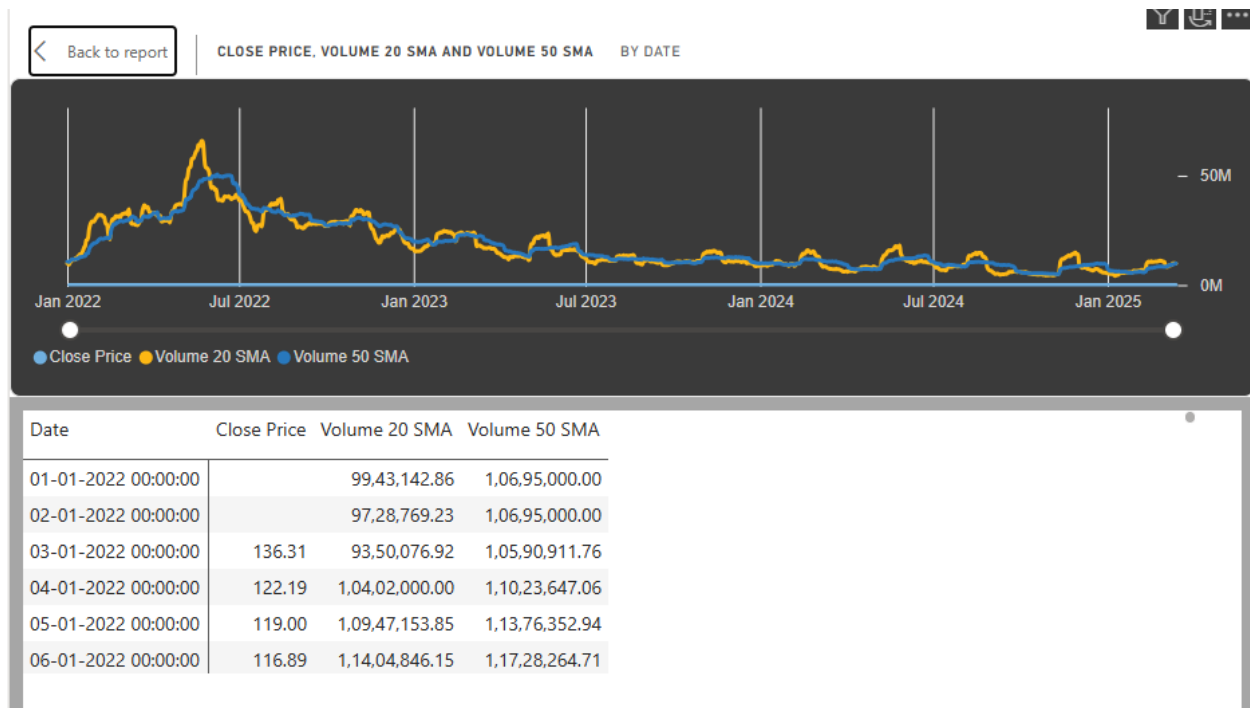
A line and stacked column chart was created to analyze the stock price and moving-average trends over time.

The visualization contains:

- Close Price
- Volume 20 SMA
- Volume 50 SMA
- Date

The date axis was configured as a continuous timeline, allowing the stock trend to be viewed across the complete period.

A zoom slider was also included to allow closer examination of specific periods



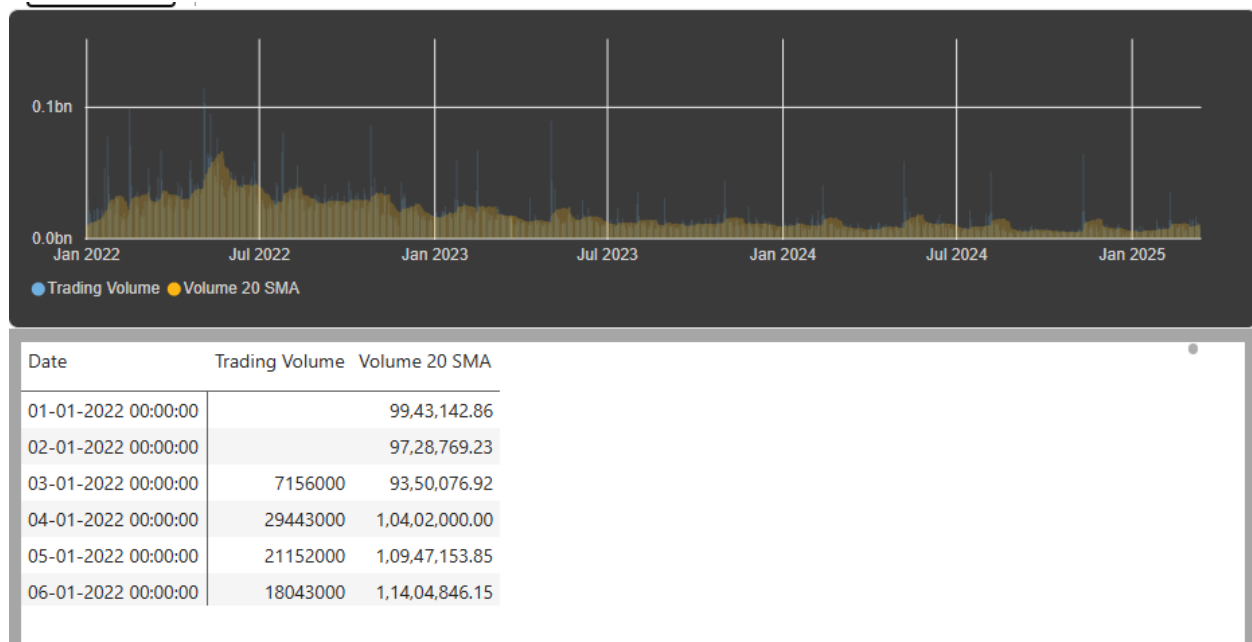
8. Trading Volume Analysis

A separate visualization was created to analyze trading volume over time.

The chart contains:

- Trading Volume
- Volume 20 SMA
- Date

The visualization helps identify periods of high and low trading activity.



9. Moving Average Crossover Signal

A crossover signal was created by comparing the 20-Day SMA with the 50-Day SMA.

The measure categorizes the relationship between the two moving averages as:

- **Bullish (20 > 50)**
- **Bearish (20 < 50)**

The DAX measure used was:

MA Crossover Signal =

IF(

ISBLANK([Volume 20 SMA]) ||

ISBLANK([Volume 50 SMA]),

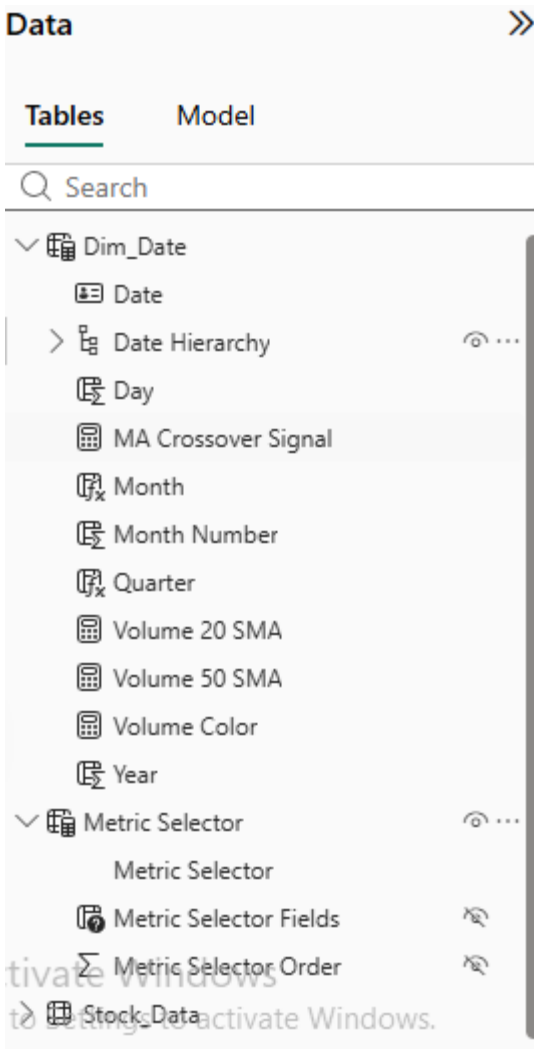
BLANK(),

IF(

[Volume 20 SMA] > [Volume 50 SMA],

"Bullish (20 > 50)",

"Bearish (20 < 50)"



10. Volume Color Measure

A Volume Color measure was created to identify unusually high trading volume.

The measure compares the current trading volume with the 20-Day SMA.

Volume Color =

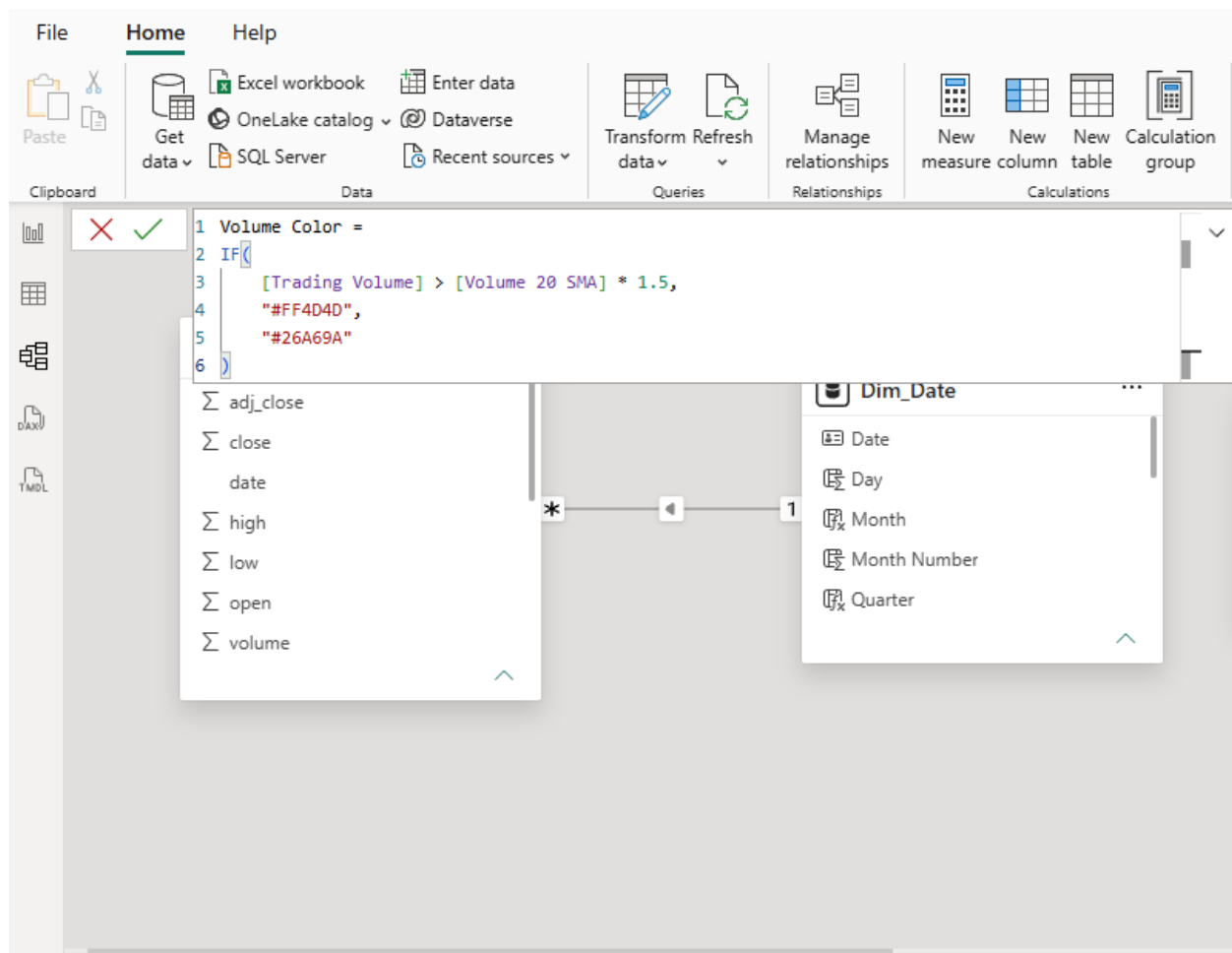
IF(

[Trading Volume] > [Volume 20 SMA] * 1.5,

"#FF4D4D",

"#26A69A"

)



11. Metric Selector Using Field Parameters

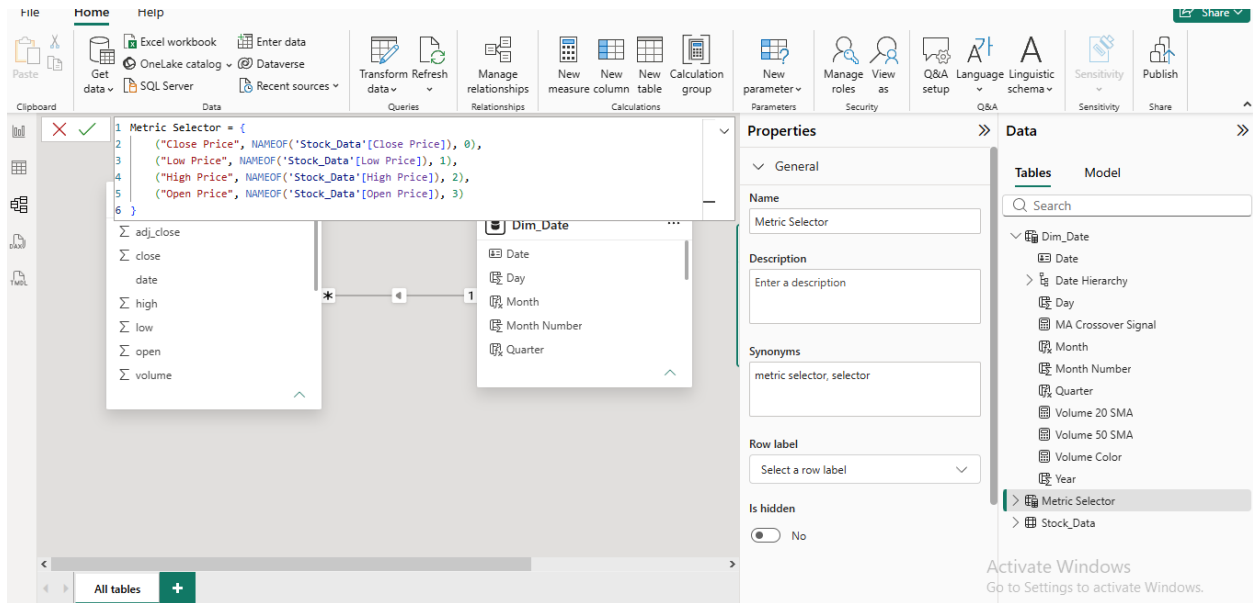
A **Metric Selector** was created using Power BI Field Parameters.

The parameter allows the user to select different stock-price metrics dynamically.

The selected fields include:

- Close Price
- Low Price
- High Price
- Open Price

The DAX generated for the parameter contains these fields and allows them to be selected dynamically in a visualization.



12. Interactive Analysis

The dashboard was designed to support interactive analysis.

Users can analyze the data using:

- Date selection
- Date range filtering
- Metric selection
- Moving-average comparisons
- Trading-volume analysis
- Stock-price trends

These interactive features make it easier to explore different periods and financial indicators without creating separate visualizations for every metric.

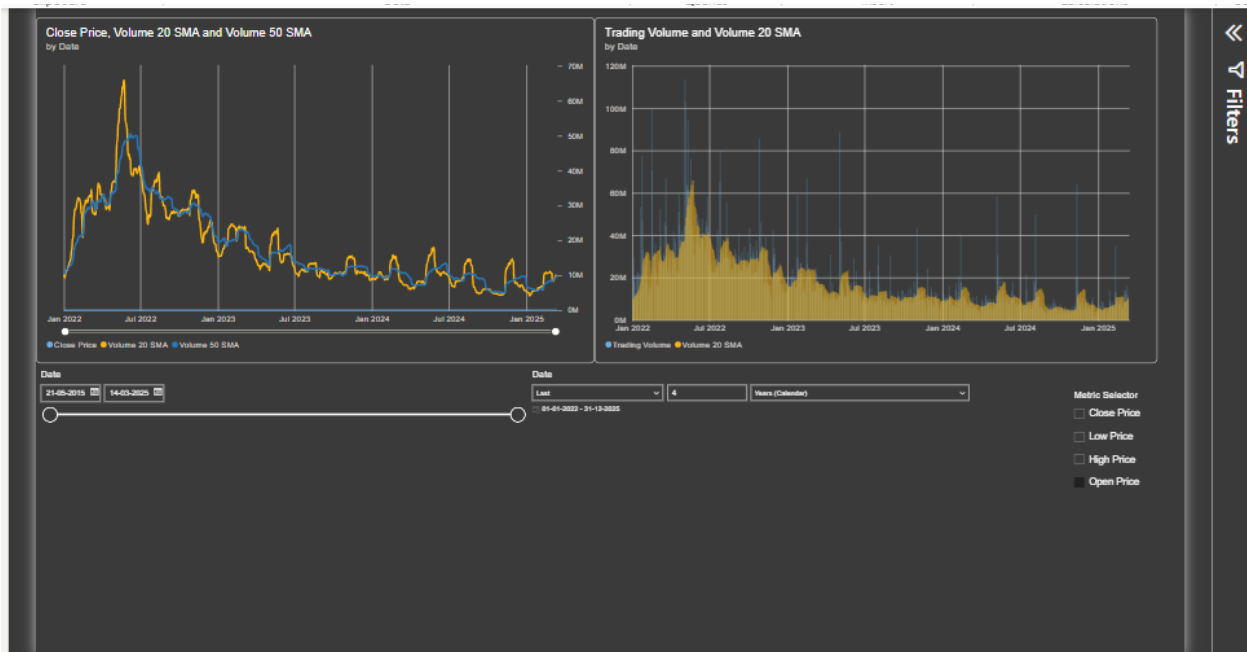
13. Dashboard Design

A dark theme was used for the dashboard to provide a modern financial-analysis appearance.

The dashboard contains KPI cards, stock price charts, trading-volume charts and interactive controls.

The dark background provides contrast for the charts and makes the financial indicators easier to distinguish.

The use of different colors also helps separate the moving-average lines and identify volume-related changes.



14. Observations

From the completed Power BI analysis, the following observations can be made:

- The Shopify stock price changes considerably over the analyzed period.
- The Close Price shows noticeable fluctuations over time.
- The 20-Day SMA provides a shorter-term view of trading activity.
- The 50-Day SMA provides a longer-term trend for comparison.
- Trading volume varies significantly across different dates.
- Certain periods show noticeable increases in trading volume.
- Comparing the 20-Day and 50-Day SMA helps identify bullish and bearish conditions.
- The Metric Selector allows different price measures to be analyzed interactively.

16. Final Deliverables

Requirement	Completed
Shopify stock data analyzed	✓
20-Day SMA created	✓
50-Day SMA created	✓
DATESINPERIOD() used	✓
MA Crossover Signal created	✓
Bullish/Bearish condition created	✓
Price trend visualization	✓
Trading volume visualization	✓
Volume Color measure	✓
Date filtering	✓
Metric Selector	✓
Field Parameter	✓
Interactive dashboard	✓

15. Conclusion

The Week 4 project demonstrates how Power BI can be used for exploratory analysis of financial time-series data.

Moving averages were created using DAX and DATESINPERIOD() to analyze short-term and long-term trading-volume trends. A moving-average crossover signal was also created to classify the relationship between the 20-Day and 50-Day SMA.

The project also includes price-trend visualization, trading-volume analysis, a volume color measure, interactive metric selection and dashboard-based exploration.

Overall, the Power BI dashboard provides an interactive way to examine Shopify stock price and trading activity over time.